ADAM HAZENBERG

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EXPERIENCE

Electrical Engineering Intern
Apptronik
June 2025 – August 2025

- Created a comprehensive dynamometer control software architecture to completely automate the process of performance testing both motor hardware and control firmware. This allows for the collection of bulk data that covers all relevant motor operating points from long testing routines.
- Conducted motor control studies that characterize electrical behaviors to in turn identify optimal firmware parameters.

Robotics Engineering Intern
HEBI Robotics
May 2024 – August 2024

- Performed chassis design iteration on 8 DOF mobile platform Tready to accommodate updated battery hardware, add new electronics, and make the robot IP68
- Wrote motor and actuator environment testing/benchmarking scripts in MATLAB
- Created mechanical drawings for high-cost aluminum chassis parts to be sent to the manufacturer. Incorporated GD&T standards
- Updated HRDF (HEBI Robot Configuration Format) database for new T25 Actuators

Product Team Intern
Skygauge Robotics
May 2023 – August 2023

- Aided the production of safe, reliable drone systems by designing and performing manufacturing and testing methods to ensure their successful maiden flights. This involved assembling, configuring, and certifying the entirety of the drone's hardware and software in accordance with company standards to ensure the quality of the final product.
- Performed diagnostics on faulty hardware such as mechanical components as well as custom PCBs to allow operation and/or suggest design improvements to boost future reliability.
- Investigated deviations in drone performance by isolating possible contributing factors to discover correlations that could be used to improve drone design.

Assistant Developer
Confined Space Robotics
April 2022 – August 2022

- Contributed to the development of various robotic systems in all software and hardware aspects, involving physical/mechanical robot design via modelling software, constructing robots using 3D printing technology, editing control programs, fixing damaged

PROJECTS

10-DOF 25kg Biped (Sim-to-Real RL)
Independent Engineering Project
January 2026 – Present

- Engineered a 25kg bipedal robot from scratch, achieving zero-shot sim-to-real dynamic locomotion using Proximal Policy Optimization (PPO) trained in NVIDIA Isaac Lab
- Designed and fabricated 10 high-power custom cycloidal actuators capable of up to 40NM of torque
- Bridged the sim-to-real gap by programming an action-wrapper middleware that applied domain randomization, including dynamic latency delay buffers and Exponential Moving Average (EMA) filters.

Magnus
Hebi Robotics
July 2024 – August 2024

- Engineered and assembled Magnus, a tetherless 5-DOF mobile robot designed to autonomously climb and navigate large ferromagnetic structures.
- Designed mechanically toggled magnetic pads for the base and end effector, utilizing Finite Element Analysis (FEA) to balance high-strength surface adhesion with lightweight payload requirements.
- Programmed control software in MATLAB, featuring stepping algorithms and autonomous 2D planar navigation including automated surface transition algorithms (ground, wall, ceiling) by leveraging actuator torque-sensing to dynamically detect and adapt to perpendicular slopes.
- Designed intuitive manual control systems by calculating appropriate spatial frames of reference for custom joystick mapping.
- Structured the robot's software architecture using object-oriented principles, encapsulating complex kinematic and locomotion functions into a modular class system for deployment.

EDUCATION

B.Eng. Mechatronics and Robotics Engineering

electrical components, and performing thorough testing of various systems to provide in-depth reports
- Primarily used Python and C++; enhanced previous skills while efficiently and accurately acquiring new ones

Self-Employment

Home-based CNC machine shop

June 2020 – June 2021

- Operated a successful, independent business providing custom CNC services such as routing, 3D printing, and laser engraving.
- Designed simple or complex 3D models to suit customers' needs.
- Constructed and maintained various owned machinery (3D printers, custom CNC router, laser engraver).

Queen's University

September 2021 – December 2025

SKILLS

C++ | Python | C | MATLAB | ROS2 | Linux/Ubuntu | CMake | Motor Control | Finite State Machines (FSM) | Impedance Control | Kinematics | Reinforcement Learning (PPO) | Hardware Debugging | Hardware-in-the-Loop (HIL) Testing | Robot Sensing | Electronics/PCB | SOLIDWORKS | Fusion 360 | CNC Routing & 3D Printing